

Thabang L. Mashinini-Sekgoto

LEAD DATA SCIENTIST · APPLIED AI & DATA SCIENCE

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Summary

Lead Data Scientist with 9+ years of experience across insurance, telecommunications, applied research, and higher education. I combine hands-on depth in machine learning, data engineering, distributed systems, and production AI with technical leadership, team development, and data science strategy. My career has progressed from research and analytical systems into national-scale real-time platforms and, more recently, leading the modernisation of enterprise data science capabilities across data engineering, MLOps, governance, and applied modelling.

Education

University of the Witwatersrand

Johannesburg, South Africa

PHD IN COMPUTER SCIENCE (COMMENCING 2027)

2027 –

- Proposed research explores physics-informed self-supervised learning for SAR-based flood extent mapping, with applications to data-scarce climate and insurance-risk settings.

University of the Witwatersrand

Johannesburg, South Africa

MSC IN COMPUTER SCIENCE (DISTINCTION)

2018 – 2019

- Thesis: Learning Level Set Method by Echo State Network for Image Segmentation.
- Proposed a novel spatiotemporal deep-learning formulation of variational level set segmentation, benchmarking five recurrent and convolutional architectures across four datasets.

University of the Witwatersrand

Johannesburg, South Africa

BSC HONOURS IN COMPUTER SCIENCE

2017

- Project: Wildfire Estimation Using Kernel Density Estimators.

University of the Witwatersrand

Johannesburg, South Africa

BSC IN COMPUTATIONAL & APPLIED MATHEMATICS AND ASTRONOMY

2014 – 2016

- Focused on mathematical modelling, astrophysics, simulation, and numerical methods.

Experience

ABSA Insurance

Johannesburg, South Africa

LEAD DATA SCIENTIST

Mar. 2024 – Present

- Lead the Insurance Data Science capability across analytical strategy, technical direction, delivery standards, and team development, working across business, technology, risk, architecture, and governance stakeholders.
- Modernising the capability from fragmented analytical workflows toward a cloud-first operating model on Databricks, bringing data engineering, data science, MLOps, CI/CD, monitoring, documentation, and governance into a more reproducible production lifecycle.
- Lead and remain hands-on with applied data science across telematics and behavioural analytics, customer intelligence and hyper-personalisation, fraud, underwriting, and geospatial flood and natural-catastrophe risk.
- Designed reusable data and ML engineering patterns using Spark, MLflow, Unity Catalog, Databricks Asset Bundles, automated orchestration, testing, and deployment controls, improving reliability and shortening the path from experimentation to production.
- Modernised telematics processing and analytical workflows handling millions of daily signals, reducing long-running processing cycles from months to under a day while improving data availability, operational visibility, and reproducibility.
- Developed geospatial flood and natural-catastrophe risk models across more than 230,000 insured properties, supporting underwriting and portfolio risk decisions.
- Build capability beyond individual models through Agile ways of working, technical standards, documentation, mentoring, show-and-tell sessions, and knowledge transfer across data science and data engineering.

Vodacom

Johannesburg, South Africa

SENIOR DATA SCIENTIST

Nov. 2021 – Mar. 2024

- Led a team of 10 data scientists and ML engineers building real-time analytics, optimisation, and decision-intelligence systems for national telecommunications infrastructure.
- Combined machine learning, mathematical optimisation, streaming data, and software engineering to solve network operations problems including generator optimisation, traffic forecasting, infrastructure planning, anomaly detection, resource allocation, and site prioritisation.
- Led the Smart Generator Optimisation work across more than 15,000 sites, contributing approximately R1 billion in annual operational savings through better allocation and use of mobile power infrastructure.
- Built and designed streaming systems processing tens of millions of daily telemetry and alarm events using Kafka, PyFlink, PySpark, and Kubernetes, enriching live operational data with telemetry and reference information for faster decision-making.
- Established reusable engineering practices and technical standards across the team, improving how analytical work moved from experimentation into reliable operational systems.
- Mentored practitioners and worked closely with network, engineering, operations, and business stakeholders, balancing hands-on technical delivery with team leadership and cross-functional execution.
- Received the Vodacom Star Award in 2022 for engineering contribution and impact.

IBM Research

Johannesburg, South Africa

MACHINE LEARNING RESEARCH SCIENTIST

Apr. 2020 – Nov. 2021

- Conducted applied machine-learning research in climate, environmental intelligence, remote sensing, and geospatial analytics, combining scientific experimentation with production-oriented engineering.
- Worked with large geospatial, satellite, and environmental datasets in distributed environments using TensorFlow and related ML tooling, developing forecasting and predictive models that could operate beyond single-machine research workflows.
- Deployed climate-forecasting models into IBM PAIRS Geoscope, connecting research outputs to a petabyte-scale geospatial-temporal platform used for environmental and enterprise analytics.
- Contributed to the Gauteng COVID-19 risk-index and prediction dashboard with IBM Research Africa, Wits University, and GCRO, supporting hotspot identification and healthcare-resource planning for the Gauteng Provincial Department of Health.
- Co-authored research and collaborated with international scientists and engineers, contributing across experimentation, model development, evaluation, data pipelines, and operationalisation.

Business Intelligence Services – University of the Witwatersrand

Johannesburg, South Africa

DATA SCIENTIST

Jun. 2018 – Apr. 2020

- Built analytics, reporting, and machine-learning systems for university-wide institutional planning, student success, and decision support using Python, SQL, Power BI, and statistical modelling.
- Developed student recommendation and progression analytics to help faculties and institutional teams identify where students needed support and make better academic planning decisions.
- Built a clustering-based recommendation system for the Faculty of Humanities, linking analytical work to student progression and government subsidy outcomes, with more than R2 million in annual subsidy impact attributed to the initiative.
- Facilitated analytics and machine-learning workshops for staff and students, helping institutional researchers and analysts adopt practical modelling and data-science methods in their own work.

Council for Scientific and Industrial Research (CSIR)

Pretoria, South Africa

DATA SCIENTIST AND SOFTWARE ENGINEER

Nov. 2017 – Jan. 2018

- Developed predictive analytics and operational intelligence systems supporting municipalities and public-sector decision-making initiatives.
- Built Django-based decision-support systems serving **17** municipalities — including the City of Cape Town and **16** across Gauteng — enabling operational visibility and real-time access to analytics for public-sector stakeholders.
- Applied machine learning and data engineering techniques to identify operational bottlenecks and improve service delivery planning processes.
- Collaborated within multidisciplinary teams combining software engineering, analytics, and public-sector innovation initiatives.
- Recognised for innovation in predictive modelling and enterprise solutions by [!\[\]\(b5e377c2bbbc6e867b430ea873fb0553_img.jpg\) Mail & Guardian](#), [!\[\]\(d9d46f6c0b63c3065512ba7a6ece9cf5_img.jpg\) CSIR](#), and [!\[\]\(e0188dd31b8ea5a27c0180ec5e972d61_img.jpg\) Department of Science and Technology \(DST\)](#).

ZAR X

Johannesburg, South Africa

SYSTEM ANALYST & SUPPORT

Mar. 2017 – Nov. 2017

- Provided systems analysis and operational support for a newly launched stock exchange serving **13,000+** clients, supporting trading platform stability during go-live and early operations.

Core Expertise

Leadership & Strategy	Data Science Strategy, Technical Leadership, Team Leadership, Capability Building
AI & Machine Learning	Applied Machine Learning, Generative AI, Optimisation, Forecasting, Computer Vision, Geospatial & Probabilistic Modelling
Data & Distributed Systems	Data Engineering, Distributed Computing, Batch & Streaming Architectures, Real-Time Data Systems
ML Engineering	MLOps, Production ML, ML Platforms, Model Lifecycle Management, Reproducible ML Systems
AI Engineering	LLM Systems, RAG, AI Agents, Multi-Model Systems, AI-Assisted Software Engineering
Governance	AI Governance, Model Governance, Production Controls, Responsible AI

Publications

Long-Range Seasonal Forecasting of 2m-Temperature with Machine Learning

NEURIPS 2020

2020

- Published research focused on probabilistic forecasting and machine learning methods for climate intelligence systems.

ML-based Probabilistic Prediction of 2m Temperature and Total Precipitation

EGU 2022

2022

- Research focused on machine learning approaches for probabilistic climate forecasting.

Mine Worker Threshold Shift Estimation via Optimization Algorithms for Deep Recurrent Neural Networks

IFAC MMM 2019

2019

- Research focused on optimisation and recurrent neural networks in mining applications.

Selected Open Source & Technical Work

Ubunye Engine

FOUNDER & MAINTAINER

2024 – Present

- Created and maintain an open-source data and ML engineering framework for building reusable workflows across local, distributed, containerised, and cloud environments.
- Designed its architecture around configuration-driven pipelines, extensible execution backends, reproducibility, testing, and production deployment—bringing together lessons from recurring data and ML engineering problems encountered in enterprise environments.
- Published the framework with documentation and practical examples for practitioners to inspect, extend, and reuse.
- [Documentation](#) [GitHub](#)

tfilterspy

CREATOR & MAINTAINER

2023 – Present

- Created an open-source Bayesian filtering and state-estimation library for noisy telemetry and time-series problems encountered across telecommunications and insurance.
- Translated Kalman, Particle, and related probabilistic filtering methods into reusable Python abstractions with support for scalable execution, bridging mathematical modelling, software engineering, and production data science.
- [Documentation](#) [GitHub](#)

FabAcademic Unfiltered

AI EDUCATOR & CO-HOST

2026 – Present

- Co-host a global AI education and knowledge-sharing series with Prof. Mamokgethi Phakeng, making practical AI accessible to audiences across countries and backgrounds, with sessions reaching audiences of over **300,000** people.
- Translate complex AI concepts into practical demonstrations and accessible conversations around learning, research, career development, and building with AI.
- [Watch on YouTube](#) [Talks & Media](#)

Recognition & Leadership

2022 **Vodacom Star Award**, Recognition for engineering contribution and impact

2014–2017 **Co-Founder, Wits Astronomy Club**, STEM outreach and astronomy public engagement

*Vodacom
University of the
Witwatersrand*

References

Dr. **Etienne Vos (IBM Research Manager)**,  Etienne.vos@ibm.com

IBM Research

Dr. **Jacobus Du Toit (Machine Learning Lead)**,  jaco.dutoit@vodacom.co.za

Vodacom

Dr. **Akram Zaytar (Senior Research Scientist)**,  akramzaytar@microsoft.com

*IBM/Microsoft
Research*

Mrs. **Harshila Dulabh (Team Lead)**,  Harshila.Dulabh@wits.ac.za

Wits University (BIS)